ReTraceVLA: Retracing a Demonstrated Route with a Memory-Conditioned Action Expert

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Abstract

RouteStick requires a robot to watch a video of a stick circling a set of posts and then repeat that path. The path is absent from the live view, so the policy has to retain the demonstration. MemoryVLA, a perceptual-cognitive memory policy, averages a success rate of 0.01 on this task over three seeds. This note introduces ReTraceVLA, a $\pi_{0.5}$ policy that encodes the demonstration with spatial pooling and temporal attention and conditions a flow-matching action expert on that trace, together with a short predicted waypoint. At checkpoint 9999, closed-loop evaluation on the RoboMME test split succeeds on 4 of 50 episodes (0.08) at seed 42. All four successes are easy episodes. This is an interim result: the remaining RoboMME suites are still in progress, and the current checkpoint admits the trace into the controller only through a near-zero residual gate.

1 Introduction

Long-horizon manipulation is often non-Markovian. The same tabletop can demand different actions once a count, a hidden object, a referred target, or a demonstrated motion has left the current frame. RoboMME organizes these demands into four suites—Counting, Permanence, Reference, and Imitation—covering temporal, spatial, object, and procedural memory [1]. RouteStick sits in the Imitation suite. The robot first watches a video in which a stick travels around posts, then must reproduce that circling path with a stick mounted on the gripper. Success depends on which side of the posts the demonstration used. A policy that sees only the current image has no basis for that choice.

MemoryVLA addresses temporal dependence with a perceptual-cognitive memory bank: a vision-language model writes working-memory tokens, a bank stores perceptual detail and semantic gist, and a diffusion action expert is conditioned on retrieved entries [2]. On the RoboMME evaluation used here, that baseline is uneven across suites and is near floor on RouteStick (average 0.01; seed 42 is 0.00). A generic memory of past interaction is a weak carrier for a specific demonstrated trajectory.

ReTraceVLA keeps the demonstration as an explicit visual trace and feeds it to the action expert alongside the live observation and the language instruction. The evaluated checkpoint also predicts a short relative waypoint from that trace and appends the waypoint as extra tokens. We report the first closed-loop measurement of this checkpoint. It is a progress note, written while training and the rest of the benchmark evaluation continue.

2 Method

ReTraceVLA fine-tunes $\pi_{0.5}$ [3]. The backbone is a PaliGemma vision-language model with a SigLIP image encoder and a flow-matching Gemma action expert. Figure 1 shows the two streams that condition the expert.

Demonstration trace. Front and wrist frames from the reference video are passed through SigLIP. Each frame is pooled into a 2×2 grid of spatial bins, so the tokens correspond to different image regions. A two-layer temporal transformer, with sinusoidal positional encodings, mixes the clip along time inside each view and bin. A linear map sends the result to the memory width (256). The clip is a uniform subsample of 16 demonstration frames and is encoded once per episode. It is a reference, separate from the frames observed while the arm is moving.

Live context. The current front and wrist images and the task instruction go through SigLIP and the PaliGemma transformer, as in $\pi_{0.5}$. On RouteStick the instruction is to watch the video and then navigate with the stick attached to the robot. Language selects the task. It does not name the circling side or a waypoint.

Action expert. The demonstration tokens and the live context condition a flow-matching action expert (width 1024, depth 18). Memory is read by residual cross-attention at expert layers 3, 7, 11, and 15. The residual is multiplied by $\tanh(\alpha)$, where the scalar gate α is initialized at 10^{-3}. The expert predicts a 20-step chunk of absolute 8-D joint positions (seven arm joints and the gripper).

Waypoint tokens. The checkpoint evaluated below adds a route head on the demonstration tokens. The head attends over the trace, reads the current pooled view and proprioception, and predicts an ordered 8-step relative TCP displacement. Those coordinates are linearly embedded, given a sinusoidal time code, and concatenated with the demonstration memory. At test time the head's own prediction is used. Training supervises it with a Huber loss on privileged relative TCP from the demonstrations ($\lambda_{\text{route}} = 1$, Huber threshold 0.02). A completion head is trained with a masked binary cross-entropy on a segment-done label ($\lambda_{\text{done}} = 0.1$) and does not itself move the robot. The flow-matching loss on the joint chunk is the primary objective.

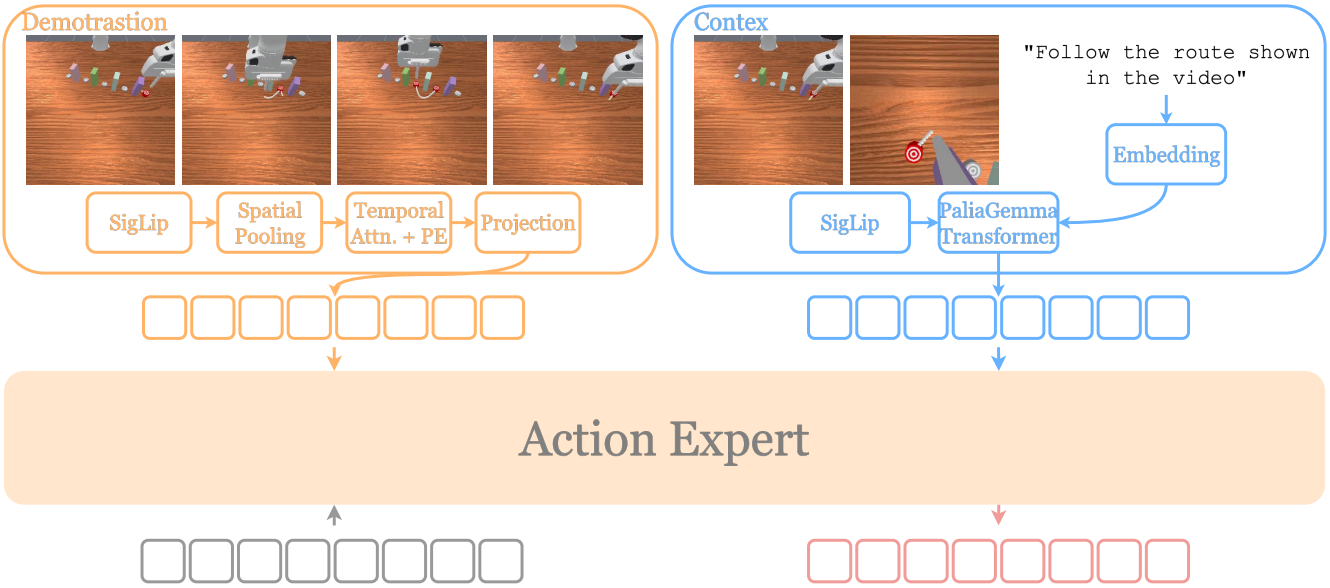


Figure 1: ReTraceVLA. The demonstration clip is encoded by SigLIP, reduced by spatial pooling, mixed by temporal attention with positional encodings, and projected to memory tokens. The live views and the instruction are encoded by SigLIP and PaliGemma. Both streams condition the action expert, which denoises a joint-position chunk. The evaluated checkpoint also appends an 8-step predicted waypoint, described in the text.

Optimization. Weights start from the public $\pi_{0.5}$ checkpoint. Training uses the RouteStick training demonstrations, AdamW with gradient clipping at 1, peak learning rate 5×10^{-5}, and 1,000 warmup steps. The run is 10,000 steps at batch size 4 on two GPUs with FSDP. We evaluate step 9999.

3 Experiments

3.1 Protocol

Evaluation follows the RoboMME closed-loop protocol on the RouteStick test split: 50 episodes, seed 42, binary task success. Episodes are labeled easy, medium, or hard by the benchmark (26, 12, and 12 episodes). The policy is served from checkpoint 9999 and receives the demonstration through the history buffer at the start of each episode. We compare against MemoryVLA numbers on the same benchmark, measured at seeds 7, 42, and 0. ReTraceVLA has been run on RouteStick only, and only at seed 42. Table 1 is included so the baseline's behavior on the other suites is visible while those runs are still outstanding.

3.2 MemoryVLA on RoboMME

Table 1 lists MemoryVLA success rates. The best suite averages are PickXtimes (0.17), VideoUnmask (0.16), MoveCube (0.14), and VideoPlaceButton (0.13). StopCube, InsertPeg, and VideoRepick are 0.00 at every seed. RouteStick is 0.02, 0.00, and 0.02 at seeds 7, 42, and 0, for an average of 0.01. Procedural imitation of a circling path is among the weakest cells in the table, together with the time-critical and precise-insertion tasks.

3.3 ReTraceVLA on RouteStick

Table 2 compares the two policies on RouteStick. ReTraceVLA at checkpoint 9999 succeeds on episodes 9, 29, 40, and 41, which is 4 of 50 episodes (0.08). On the matched seed, MemoryVLA succeeds on none. The comparison is a single seed against a three-seed baseline, so we treat 0.08 as an early measurement of this checkpoint, not as a suite-level ranking.

Table 3 splits the same 50 episodes by difficulty. The four successes are all easy. Medium and hard episodes contribute no successes. Even on easy episodes the rate is 4 of 26. Most demonstrations are still not reproduced.

Training diagnostics at step 9950 are consistent with a trace that is fit in the auxiliary head and weakly used by the controller. The flow-matching loss is 0.013, and the route head's mean waypoint error on the training batch is 0.019. The adapter gate that scales memory cross-attention remains 0.0009, against an initialization of 10^{-3}.

4 Current limitations

The result above is a single checkpoint on a single task and a single seed. Seeds 7 and 0 have not been run, and Counting, Permanence, and Reference have not been run. A 0.08 success rate leaves 46 of 50 test episodes failed, including every medium and hard episode.

The route head can match privileged training waypoints while the action expert barely reads them. Memory reaches the expert only as a residual, and that residual stays scaled by about 10^{-3} through the end of this run. Raising the gate, or otherwise giving the demonstration a stronger path into the expert, is the immediate change under development. A matched run of the same backbone with the demonstration stream removed is

Table 1: MemoryVLA success rate on RoboMME. Each entry is one seed. Avg is the unweighted mean of seeds 7, 42, and 0.

Suite	Task	Seed 7	Seed 42	Seed 0	Avg
Counting	BinFill	0.12	0.10	0.08	0.10
	PickXtimes	0.06	0.24	0.22	0.17
	SwingXtimes	0.00	0.02	0.02	0.01
	StopCube	0.00	0.00	0.00	0.00
Permanence	VideoUnmask	0.10	0.18	0.20	0.16
	VideoUnmaskSwap	0.04	0.04	0.06	0.05
	ButtonUnmask	0.04	0.06	0.14	0.08
	ButtonUnmaskSwap	0.04	0.04	0.06	0.05
Reference	PickHighlight	0.04	0.10	0.08	0.07
	VideoRepick	0.00	0.00	0.00	0.00
	VideoPlaceButton	0.12	0.16	0.12	0.13
	VideoPlaceOrder	0.08	0.00	0.04	0.04
Imitation	MoveCube	0.12	0.18	0.12	0.14
	InsertPeg	0.00	0.00	0.00	0.00
	PatternLock	0.12	0.10	0.14	0.12
	RouteStick	0.02	0.00	0.02	0.01

Table 2: RouteStick success. ReTraceVLA is checkpoint 9999. MemoryVLA is the three-seed baseline from Table 1.

Method	Seed	Success
MemoryVLA	7	0.02
MemoryVLA	42	0.00
MemoryVLA	0	0.02
MemoryVLA	mean	0.01
ReTraceVLA	42	0.08 (4/50)

Table 3: ReTraceVLA by RouteStick difficulty, seed 42, checkpoint 9999.

Difficulty	Successes	Rate
Easy	4/26	0.15
Medium	0/12	0.00
Hard	0/12	0.00
All	4/50	0.08

also still missing, so the 4 successes are not yet isolated from ordinary visuomotor behavior on the easy split.

Success here is the benchmark's binary flag. It does not report whether a failed episode chose the wrong circling side, missed a post, or left the path early. Those failure modes matter for the next iteration and are not in this note.

5 Conclusion

ReTraceVLA encodes a RouteStick demonstration as a spatiotemporal trace, predicts a short waypoint from it, and conditions a $\pi_{0.5}$ action expert on both. Checkpoint 9999 succeeds on 4 of 50 test episodes at seed 42, against a MemoryVLA average of 0.01 and a seed-42 rate of 0.00. The successes are confined to easy episodes, and the memory residual that should carry the trace is still near its initialization. The rest of the benchmark, the remaining seeds, and a stronger coupling from the trace into the action expert are the continuation of this work.

References

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[3] Physical Intelligence. $\pi_{0.5}$: A vision-language-action model with open-world generalization. *arXiv preprint arXiv:2504.16054*, 2025.